Congfeng Cao

Updated June 29, 2026

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About me

I am a final-year PhD student at the University of Amsterdam, working on LLM post-training and evaluation, RAG, agentic systems, and graph-based methods. Before my PhD, I worked in industry as a machine learning researcher and team leader, building NLP, knowledge graph, text mining, and question-answering systems for real-world applications. My technical experience includes Python, PyTorch, TensorFlow, LLM fine-tuning, graph neural networks, retrieval systems, agentic workflows, full-stack data science, and databases.

Skills

Proficient (10 years): Python, PyTorch, and TensorFlow.
Databases (10 years): SQL (MySQL), NoSQL (Neo4J, MongoDB), PostgreSQL.
Data science (10 years): Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn.
Familiar with agent frameworks (LangChain, CrewAI and OpenAI Agents SDK), full-stack data science, C/C++, and Java.

Research experience

NLP Unit, Institute for Logic, Language and Computation, UvA
Promoter: Khalil Sima'an, Supervisor: Jelke Bloem — October 2022 - Present
LLM post-training and evaluation, and the coordination and alignment between language models and graph models.

University of Chinese Academy of Sciences (UCAS)
Supervisor: Junyong Fang — October 2014 - July 2017
Analyzing multi-band remote sensing images using deep learning and computer vision techniques.

Medical College, Peking University
Part-time Researcher — 2018 - 2019
Information retrieval and question answering for Chinese medicine literature.

Industry experience

Kunlun Digital Technology Co.,Ltd. (or CNPC Richfit)
Senior Machine Learning Researcher — January 2021 - October 2022
Team Leader in Conversation Service Group. Responsible for building an automated customer question-answering system, leading the team, and managing projects. Service robots equipped with the QA system [Video] [Description].

Tsinghua Tongfang Co., Ltd.
Machine Learning Researcher — July 2017 - January 2021
Team Leader in Machine Learning Group. Responsible for research on natural language processing, knowledge graphs, and text mining algorithms; applying these algorithms in projects; and leading the team while managing projects.

Selected Publications

Fusion Training for Mathematical Generalization in Large Language Models (*ACL 2026 SRW*)| Code
Congfeng Cao*, Pengyu Zhang, Jelke Bloem.

Contribution: *Reveals thinking and non-thinking training modes interfere, and mixed training gives the best balance.*
Technical: *Qwen3-4B base model, Supervised Fine-Tuning (SFT), GSM8K, OpenMathReasoning, thinking/non-thinking data ratios, T-NT/NT-T/Mix schedules.*

Graph-TempCZ: A Graph Representation of Software Mentions for Predicting Software Usage in Scientific Publications. (*LREC 2026*)| Code
Congfeng Cao*, Pengyu Zhang, Jelke Bloem.
Contribution: *Introduces Graph-TempCZ, a temporal graph representation of software mentions, and formulates software usage prediction in scientific publications as a link prediction task.*
Technical: *Sentence-BERT, XGBoost, GraphSAGE, temporal graph evaluation, CZ Software Mentions, and Europe PMC metadata.*

How Aligned Are Unimodal Language and Graph Encodings of Chemical Molecules? (*IJCNLP-AACL Main 2025*)| Code
Congfeng Cao*, Zhi Zhang, Jelke Bloem, Khalil Sima'an.
Contribution: *Studies whether molecular graph and language encoders are inherently aligned before force-alignment training, and introduces the U2A benchmark to evaluate this alignment.*
Technical: *MolCLR, MolFormer, SciBERT, SentenceBERT, GPT, Debiased CKA, U2A benchmark, ChEBI-20, RDKit, and MAYGEN.*

Grants

NWO Compute Projects 2024, 2025, 2026

UCAS Academic Scholarship 2015, 2016, 2017

Education

University of Amsterdam Amsterdam, Netherlands
PhD in Computer Science October 2022 - Present

University of Chinese Academy of Sciences Beijing, China
MEng in Signal and Information Processing October 2014 - July 2017

Presentations

ACL 2026, LREC 2026, IJCNLP-AACL 2025, EMNLP 2025, CLIN 2025, CLIN 2024, ELLIS 2024.

Course & Training

Knowledge Graphs (2026), IBM RAG and Agentic AI Professional Certificate (2025), Agentic AI (2025), Fine-tuning & RL for LLMs (2025), Generative AI with Large Language Models (2025), Machine Learning and Optimisation (2024), LxMLS (2024), LOGML (2025), Human-Centred AI (2023)

Under Review

SoftwareCiter: Agentic Generation of Verifiable Software Citations with a Metadata-Aware Benchmark
Congfeng Cao*, Yixian Shen, Kayle Verhiel, Jelke Bloem.
Contribution: *Proposes a multi-agent framework for generating complete and verifiable software citations from incomplete software mentions in publications*
Technical: *LLM agents, LangChain, OpenAI API, publication parsing, metadata extraction, web searching, FORCE11 citation principle, and metadata verification.*

Molecule Generation through Reasoning with Large Language Models
Congfeng Cao*, Di Wu*, Jiahong Huang, Jelke Bloem, Khalil Sima'an.

Contribution: *Frames molecule generation as reasoning-based SMILES generation and introduces MolCoT-4K and MolGeneration.*
Technical: *Qwen2.5-7B base model, curated CoT data, SFT, GRPO RL, RDKit validity checking, exact-match rewards, Pass@N evaluation, OpenAI API, Gemini API, Claude API, and DeepSeek API.*

Rethinking Feature Augmentation in Graph Contrastive Learning
Congfeng Cao*, Pengyu Zhang, Zeyu Zhang, Jelke Bloem.
Contribution: *Introduces Random Feature Masking (RFM), a simple graph contrastive learning augmentation that improves node classification performance and robustness, especially at high masking rates.*
Technical: *Graph contrastive learning, feature augmentation, GRACE, GraphCL, CCA-SSG, node classification, and uniformity analysis.*

A Graph-Based Analysis of Software Usage in Scientific Publications
Congfeng Cao*, Na Li, Kristy James, Jelke Bloem.
Contribution: *Shows software forms structured, domain-specific ecosystems rather than isolated frequency patterns.*
Technical: *Graph-CZ, Software-CZ, Publication-CZ, temporal analysis, community detection, and cross-community interaction analysis.*

Collaboration

NeuroAda: Activating Each Neuron's Potential for Parameter-Efficient Fine-Tuning (EMNLP 2025)
Zhi Zhang*, Yixian Shen*, **Congfeng Cao**, Ekaterina Shutova.

Cycle: Cross-year contrastive learning in entity-linking (CIKM 2024)
Pengyu Zhang*, **Congfeng Cao**, Klim Zaporojets, Paul Groth.

TIGER: temporally improved graph entity linker (ECAI 2024)
Pengyu Zhang*, **Congfeng Cao**, Paul Groth.

Languages

Chinese (native), English (work language), Dutch (Beginner)

Other interests

Walking and Hiking.